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Redistribution layer having a sideview zig-zag profile

Abstract

A semiconductor device package includes a semiconductor device and an electrically conductive pad disposed in contact with a surface of the semiconductor device. The semiconductor device package further includes a redistribution layer (RDL) formed over the electrically conductive pad and the surface of the semiconductor device, and an electrical connector disposed over and electrically coupled to the RDL. The RDL includes a first passivation layer disposed over a surface of the semiconductor device and the electrically conductive pad, and further includes an RDL trace. The RDL trace includes a first portion in contact with the electrically conductive pad, a second portion in contact with one of the electrical connector or an underlying metallization layer in contact with the electrical connector, and a third portion having a non-planar and undulating configuration relative to the surface of the semiconductor device.

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Background/Summary

BACKGROUND

(1) Wire bonding often is used to electrically couple a semiconductor device to a packaging substrate. However, the wire connections are typically limited to the outer perimeter of the semiconductor device, which limits input/output (I/O) density. Also, the presence of wire bonds beyond the semiconductor device increases the area necessary for the final device package. Therefore, layers of dielectric material embedded with electrically conductive metal lines and vias, referred to as redistribution layers (RDL), are used to reroute wire bond connections from the edge to the center of the semiconductor device. After the RDL is formed, the device packaging process flow can continue using, for example, under bump metallization (UBM) instead of conventional wire bonding. In a UBM process, the metallization is deposited on the top metal level of the semiconductor device and solder metal is used to directly connect the semiconductor device to the packaging substrate. The connection points between the device and the packaging substrate are distributed over the entire top surface of the chip, which increases the I/O density by using a higher percentage of the semiconductor device surface area for connection to the packaging substrate. As such, RDLs can be used to maintain existing package designs while accommodating the smaller die produced by semiconductor manufacturers transitioning to advanced technology nodes.

SUMMARY OF EMBODIMENTS

(2) In accordance with one aspect, a semiconductor device package includes a semiconductor device, a first passivation layer disposed over a surface of the semiconductor device, and a redistribution layer (RDL) trace disposed in physical contact with the first passivation layer. The RDL trace includes a non-planar configuration relative to the surface of the semiconductor device, wherein the non-planar configuration includes a first area of the RDL trace having a first height and a second area of the RDL trace having a second height different than the first height.

(3) The semiconductor device package may include one or more of the following features, individually or in combination: the non-planar configuration includes at least one peak and at least one valley in the RDL trace; an electrically conductive pad disposed below the first passivation layer and in electrical contact with the semiconductor device and the RDL trace; a second passivation layer disposed above and in physical contact with the RDL trace; an electrical connector disposed above and electrically coupled to the RDL trace; a metallization layer disposed between the RDL trace and the electrical connector, wherein a first surface of the metallization layer physically contacts the RDL trace, and a second surface of the metallization layer physically contacts the electrical connector; a third passivation layer in physical contact with the surface of the semiconductor device and a bottom surface of the first passivation layer, and further in physical contact with the electrically conductive pad; wherein the RDL trace includes a first portion in physical contact with the electrically conductive pad, a second portion in physical contact with the electrical connector, and a third portion comprising the non-planar configuration, wherein the third portion is disposed between the first portion and the second portion of the RDL trace; or wherein a first surface of the third portion of the RDL trace transitioning from a first area to a second area of the RDL trace includes is sloped such that a first obtuse angle is formed between the first area and the second area, and wherein a second surface of the third portion of the RDL trace transitioning

from the second area to a third area of the RDL trace is also sloped such that a second obtuse angle is formed between the second area and third area.

(4) In accordance with another aspect, a semiconductor device package includes a semiconductor device, an electrically conductive pad disposed in electrical contact with the semiconductor device, a redistribution layer (RDL) formed over the electrically conductive pad, and an electrical connector disposed over and electrically coupled to the RDL. The RDL includes a first passivation layer disposed over a surface of the semiconductor device and the electrically conductive pad, and RDL trace having a first portion in physical contact with the electrically conductive pad, a second portion in physical contact with one of the electrical connector or an underlying metallization layer in physical contact with the electrical connector, and a third portion including a non-planar and undulating configuration relative to the surface of the semiconductor device.

(5) The semiconductor device package may include one or more of the following features, individually or in combination: the non-planar and undulating configuration includes at least one peak and at least one valley in the RDL trace; wherein the at least one peak is disposed in at least a first area of the third portion of the RDL trace and the at least one valley is disposed in at least a second area of the third portion of the RDL trace, and wherein a surface of the RDL trace transitioning from the first area to the second area is sloped such that an obtuse angle is formed between the first area and the second area; wherein the semiconductor device package further includes a second passivation layer disposed on and in physical contact with the RDL trace; wherein the metallization layer is disposed within an opening formed in the second passivation layer; wherein the metallization layer is disposed between the RDL trace and the electrical connector, wherein a first surface of the metallization layer physically contacts the RDL trace, and a second surface of the metallization layer physically contacts the electrical connector; or another passivation layer in physical contact with the surface of the semiconductor device and a bottom surface of the first passivation layer, and further in physical contact with the electrically conductive pad.

(6) In accordance with yet another aspect, a method of fabricating a semiconductor device package includes forming a first passivation layer over a surface of an underlying semiconductor device; patterning the first passivation layer, wherein the patterning forms a first non-planar and undulating configuration in a portion of the first passivation layer relative to the surface of the semiconductor device; and forming a redistribution layer (RDL) trace in physical contact with the patterned first passivation layer, wherein the RDL trace includes a second non-planar and undulating configuration relative to the surface of the semiconductor device and conforming to the first non-planar and undulating configuration.

(7) The method of fabricating the semiconductor device package may include one or more of the following features, individually or in combination: patterning the first passivation layer using at least one of a grayscale mask or a multi-exposure process; wherein patterning the first passivation layer forms a trench in the first passivation layer that exposes an electrically conductive pad in contact with the semiconductor device. wherein forming the RDL trace includes forming a first portion of the RDL trace in the trench and in contact with electrically conductive pad, forming a second portion of the RDL trace opposite the first portion of the RDL trace, and forming a third portion of the RDL trace between the first portion and the second portion, wherein the third portion of the RDL trace comprises the second non-planar and undulating configuration; forming a second passivation layer on and in contact with the RDL trace, forming a trench in the second passivation layer exposing a portion of the RDL trace, and forming an electrical connector in electrical contact with the exposed portion of the RDL trace; or wherein the first non-planar and undulating configuration and the second non-planar and undulating configuration each comprise at least one peak and at least one valley.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is better understood, and its numerous features and advantages made apparent to those skilled in the art, by referencing the accompanying drawings. The use of the same reference symbols in different drawings indicates similar or identical items.
- (2) FIG. 1 is a top view of a semiconductor device package and a cross-section view **102** of a portion of the semiconductor device package implementing a non-planar redistribution layer configuration in accordance with one or more embodiments.
- (3) FIG. 2 illustrates a method for fabricating a semiconductor package implementing a non-planar redistribution layer including processes for forming a passivation layer and a first re-passivation layer in accordance with one or more embodiments.
- (4) FIG. 3 illustrates a continuation of the method started in FIG. 2 including processes for forming a hilling topography in the first re-passivation layer in accordance with one or more embodiments.
- (5) FIG. 4 illustrates a continuation of the method illustrated in FIG. 2 and FIG. 3 including processes for forming redistribution layer having a non-planar configuration in accordance with one or more embodiments.
- (6) FIG. 5 illustrates a continuation of the method illustrated in FIG. 2 to FIG. 4 including processes for patterning the redistribution layer, forming a second re-passivation layer, a seed layer for under bump metallization in accordance with one or more embodiments.
- (7) FIG. 6 illustrates a continuation of the method illustrated in FIG. 2 to FIG. 5 including processes for patterning the under bump metallization and forming an electrical bump in accordance with one or more embodiments.

DETAILED DESCRIPTION

- (8) Redistribution layers (RDLs) are commonly implemented in many semiconductor packaging technologies, such as ball grid array (BGA) packaging, chip-scale packaging (CSP), wafer-level packaging (WLP), panel-level packaging, three-dimensional (3D) packaging, system-in-a-package (SiP), and the like. RDLs interface between the underlying semiconductor device and electrical bumps (e.g., solder bumps) of the package so that signals are routed appropriately between the bumps and the active circuitry of the semiconductor device. Stated differently, RDLs reroute input/output (I/O) pads of an originally designed circuit via a wafer-level metal distribution process and a micro bump process so that the circuitry can be adapted for different packaging forms.
- (9) Conventional RDL configurations are typically planar (straight) and formed by depositing a passivation layer including an insulating dielectric material (e.g., polyimide) onto a finished back-end-of-line (BEOL) structure at wafer level, and then forming a new level of redistribution wiring/traces either in aluminum or copper metallization using standard BEOL processing. As part of the BEOL processing, a trench is usually formed in the passivation layer to expose an underlying conducting pad. The metalized RDL trace is formed and patterned in contact with the passivation layer and the exposed portion of the conducting pad. The RDL trace is then coated with another passivation layer including an insulating dielectric material (e.g., polyimide) through which the interconnect openings for solder bump connections are formed. conventional planar RDL trace configurations can be susceptible to poor adhesion resulting from, for example, the difference between the thermal properties of the passivation layers and the RDL trace, which leads to problems such as delamination and cracking. Also, conventional planar RDL trace configurations tend to be susceptible to stresses resulting from the various packaging layers and the planar configuration of the RDL trace itself, which can lead to the RDL trace becoming damaged or broken.
- (10) As such, described herein is a non-planar RDL trace configuration and method for forming the same that can mitigate the aforementioned issues associated with planar RDL trace configurations.

The non-planar RDL trace configuration comprises a hilling topography, which includes one or more peaks having a first height and one or more valleys having a second height different from the first height. The hilling topography is formed, for example, using a gradient grayscale mask or a multi-exposure process that forms a peak and valley pattern in a passivation layer overlying a semiconductor device. This process also forms a trench/opening in the passivation layer exposing an electrically conductive pad(s), such as a bonding pad, underlying the passivation layer. An electrically conductive material, such as copper, titanium, tungsten, aluminum, or the like, an alloy of two or more such materials, or any suitable combination of electrically conductive materials is deposited over the patterned passivation layer and fills the trench. The electrically conductive material is patterned to form an RDL trace including a first portion in contact with the bond pad and a second portion, opposite the first portion, configured to receive either an electrical bump (e.g., solder bump/ball) or a combination of UBM and an electrical bump. As the RDL trace conforms to the patterned passivation layer, a third portion of the RDL trace, which is disposed between the first and second portions of the RDL trace, comprises a non-planar configuration having a peak and valley pattern. In one or more embodiments, the non-planar configuration of the RDL trace is characterized as a “zig-zag” or “folding” pattern when taken across the longitudinal axis of the RDL trace along a plane orthogonal to the underlying surface of the semiconductor device. The non-planar configuration of the RDL trace provides an increased stress buffer along the stretching direction (i.e., longitudinal axis) of the RDL trace compared to a conventional planar (straight) RDL trace configuration. The increased stress buffer can avoid, or at least mitigate, rapid stress accumulation in the RDL trace. Moreover, the non-planar configuration of the RDL trace can increase the contact area between the RDL trace and underlying/overlying passivation layers compared to the conventional planar RDL trace configuration. The increased contact area can provide for improved interlayer adhesion between the RDL trace and underlying/overlying passivation layers. Also, the “zig-zag” pattern of the non-planar RDL trace configuration can increase the electroplating charging path when forming the RDL trace compared to the conventional planar configuration, which can improve plating thickness uniformity.

(11) Note that in the following, certain orientational terms, such as top, bottom, front, back, and the like, are used in a relative sense to describe the positional relationship of various components. These terms are used with reference to the relative position of components either as shown in the corresponding figure or as used by convention in the art and are not intended to be interpreted in an absolute sense with reference to a field of gravity. Thus, for example, a surface shown in the drawing and referred to as a top surface of a component would still be properly understood as being the top surface of the component, even if, in implementation, the component was placed in an inverted position with respect to the position shown in the corresponding figure and described in this disclosure. Further, note that certain positional terms, such as co-planar or parallel, will be understood to be interpreted in the context of fabrication tolerances or industry standards. For example, co-planar shall be understood to mean co-planar within applicable tolerances as a result of one or more fabrication processes affecting the components indicated to be co-planar, or co-planar within a tolerance utilized in the appropriate industry or fabrication technology. Moreover, it will be appreciated that for simplicity and clarity of illustration, components shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the components may be exaggerated relative to other components.

(12) It also should be noted that the terms “contact”, “contacts”, “contacting”, or their equivalents refer to, for example, components, such as layers, features, or surfaces being in physical (direct) contact or indirect contact through one or more intermediate layers, features, or surfaces, or the like. Moreover, a component can be in “electrical contact” with one or more other components, either directly or indirectly through one or more intermediate components, depending on the electrical conductivity of the components' material(s). Also, the term “metal” as used herein shall be understood to refer to an elemental metal (e.g., copper (Cu) or nickel (Ni)), a metal alloy (e.g., a

copper-nickel alloy), a combination of metals, a combination of metal alloys, or a combination of one or more metals and one or more metal alloys (e.g., a copper layer with a copper-nickel-palladium plating).

(13) FIG. 1 illustrates a top view **100** of a semiconductor device package **104** and a cross-section view **102** of a portion **101** of the semiconductor device package **104** implementing a non-planar configuration **108** for an RDL trace in accordance with one or more embodiments. The cross-section view **102** is taken along the illustrated x-axis of the semiconductor device package **104** at cut line **100-100** corresponding to the longitudinal axis of the semiconductor package portion **101**. It should be understood that one or more layers shown in the cross-section view **102** are not shown in the top view **100** for illustrative purposes. It should also be understood that although the following description uses wafer-level packaging as one example of a packaging technique, other packaging techniques are applicable as well, such as panel-level packaging, other chip-scale packaging (CSP), ball grid array (BG) packaging, three-dimensional (3D) packaging, system-in-a-package (SiP), and the like. Also, although a 2P2M (two dielectric polymer re-passivation, 1 RDL metal, 1 UBM metal) packaging configuration is illustrated in the figures, the RDL configurations and techniques described herein are also applicable to other packaging configurations, such as 2P1M (two dielectric polymer re-passivation and 1 RDL metal).

(14) As illustrated, the semiconductor device package **104** comprises a semiconductor device/chip **110** and packaging layers **112** formed thereon. In one or more embodiments, the packaging layers **112** include an electrically conductive pad(s) **114** (e.g., a bonding pad), a first passivation layer **116**, an RDL **106** comprising at least a second passivation layer **118** and one or more RDL traces **120**, a third passivation layer **122**, a UBM layer(s) **124**, and an electrical connector(s)/bump(s) **126** (e.g., solder ball/bump). The second passivation layer **118** and the third passivation layer **122** are also referred herein as the “first re-passivation layer **118**” and the “second re-passivation layer **122**”, respectively. In one or more embodiments, at least one of these features comprise a single layer while, in other embodiments, one or more of these features comprise multiple layers. Also, in one or more embodiments, at least one of these features/layers, such as the UBM layer(s) **124** or one of the re-passivation layers **118** or **122**, may be omitted.

(15) The electrically conductive pad **114**, in one or more embodiments, contacts a portion of the underlying surface **128** surface of the semiconductor device **110**. In this example, the underlying surface **128** can also be referred to as the top surface **128** of the semiconductor device **110**. The electrically conductive pad **114** comprises an electrically conductive material, such as gold, aluminum, copper, the like, or a combination thereof. The passivation layer **116** is disposed in contact with the top surface **128** of the semiconductor device **110**, sidewalls of the electrically conductive pad **114**, and a portion of the top surface **205** (FIG. 2) of the electrically conductive pad **114**. The passivation layer **116** comprises a material such as nitride, oxide, polybenzoxazole (PBO), benzocyclobutene (BCB), polyimide (PI), or any other suitable passivation material. In one or more embodiments, an adhesion layer (not shown) is formed in place of, or in addition, to the passivation layer **116** to promote improved adhesion between the overlying re-passivation layer **118** and underlying semiconductor device **110**.

(16) The first re-passivation layer **118** (or other dielectric layer) is disposed in contact with a top surface **205** (FIG. 2) of the passivation layer **116**. In one or more embodiments, the first re-passivation layer **118** also contacts the top surface **205** (FIG. 2) of the electrically conductive pad **114**. A portion **130** of the first re-passivation layer **118** comprises a hilling topography **209** (FIG. 2) including peaks **132** (illustrated as peak **132-1** and peak **132-2**) and valleys **134** (illustrated as valley **134-1** and valley **134-2**). The peaks **132** are defined as having their top surface disposed closer to the top surface of the first re-passivation layer **118** than the top surface of the valleys **134**, and the valleys **134** are defined as having their top surface disposed closer to a bottom surface of the first re-passivation layer **122** than the top surface of the peaks **132**. In this example, the bottom surface of the first re-passivation layer **118** is the surface of the first re-passivation layer **118** closest

to the semiconductor device **110**. Also, the surface of the first re-passivation layer **118** that transitions from a peak **132** to a valley **134**, and vice versa, is sloped such that the height of each peak **132** is different than the height of each valley **134** relative to either the top surface or bottom surface of the first re-passivation layer **118** and these height changes are continuous across the portion **130** of the first re-passivation layer **118**. In one or more embodiments, the sloping surface forms an obtuse angle between the peak **132** and the valley **134**. The first re-passivation layer **118** comprises one or more materials such as polymer-based materials (PI, PBO, BCB, or the like), epoxy-based materials, or any other suitable insulating passivation material.

(17) The RDL trace **120** is disposed in contact with the first passivation layer **118** and the electrically conductive pad **114**. The RDL trace **120** comprises an electrically conductive material, such as copper, titanium, tungsten, aluminum, the like, or a combination thereof. In one or more embodiments, the RDL trace **120** includes or is formed on an underlying seed layer **121**, comprised of a metal, such as titanium, copper, gold, a combination thereof, or the like. In one or more embodiments, a first portion **136** of the RDL trace **120** fills a trench/opening **217** (FIG. 3) in the first re-passivation layer **118** and forms a via **138** in contact with the electrically conductive pad **114**. A second portion **140** of the RDL trace **120**, which is disposed opposite the first portion **136**, is configured to receive the electrical bump **126** and any UBM layer **124**. A third portion **142** of the RDL trace **120** is disposed between the first portion **136** and the second portion **140** of the RDL trace **120** and comprises a non-planar configuration **108** relative to the illustrated z-axis. The non-planar configuration **108** conforms to the hilling topography of the underlying first re-passivation layer **118**. As such, the non-planar configuration **108** comprises one or more peaks **144** (such as the illustrated peak **144-1** and peak **144-2**) and one or more valleys **146** (such as the illustrated valley **146-1** and valley **146-2**) forming a “zig-zag” or “folding” pattern relative to a plane orthogonal to the underlying surface **128** of the semiconductor device **110**, as shown in the cross-section view **102** of FIG. 1. The peaks **144** are defined as having their top surface disposed closer to the top surface of the RDL **106** than the top surface of the valleys **146**, and the valleys **146** are defined as having their top surface disposed closer to a bottom surface of the RDL **106** than the top surface of the peaks **144**. In this example, the bottom surface of the RDL **106** is the surface of the RDL **106** closest to the semiconductor device **110**. Also, the surface of the RDL trace **120** that transitions from a peak **144** to a valley **146**, and vice versa, is sloped such that the height of each peak **144** is different than the height of each valley **146** relative to either the top surface or bottom surface of the RDL **106** and these height changes are continuous across the third portion **142** of the RDL trace **120**. In one or more embodiments, the sloping surface forms an obtuse angle between the peak **144** and the valley **146**. An example range of the height difference (or thickness of the first re-passivation layer **118**) between a peak **146** and a valley **146** is 2 to 5 micrometers or greater. Thus, as illustrated, the RDL trace **120** may take on an undulating or quasi-sinusoidal profile relative to the illustrated Z-X plane for at least a portion of its lateral extent. In one example, the thickness of the first re-passivation layer **118** at the location of an RDL valley **146** is 5 micrometers or greater, although lesser thicknesses are applicable as well.

(18) The second re-passivation layer **122** (or other dielectric layer) is disposed in contact with exposed surfaces of the RDL trace **120**. In one or more embodiments, the second re-passivation layer **122** also contacts portions of the first re-passivation layer **118** that are not covered by the RDL trace **120**. The topography of the second re-passivation layer **122**, in one or more embodiments, conforms to the topography of the RDL trace **120**. In one or more embodiments, the second re-passivation layer **122** comprises material(s) that are the same or different than the material(s) of the first re-passivation layer **118**, such as polymer-based materials (PI, PBO, BCB, or the like), epoxy-based materials, or any other suitable insulating passivation material. In one example, the thickness of the second re-passivation layer **122** at the location of an RDL peak **144** is 5 micrometers or greater, although lesser thicknesses are applicable as well.

(19) The UBM layer **124** is disposed within and conforms to a trench **239** (FIG. 5) formed in the

second re-passivation layer **122** and contacts the second portion **140** of the RDL trace **120**. In one or more embodiments, a portion(s) **154** of the UBM layer **124** extends above/over and contacts the top surface **243** (FIG. **6**) of the second re-passivation layer **122**. The UBM layer **124** is composed of a single layer or multiple layers and comprises one or more electrically conductive materials, such as copper, titanium, gold, tungsten, nickel, chrome, or the like, an alloy of two or more such materials, or any suitable combination of electrically conductive materials. In one or more embodiments, the UBM layer **124** includes or is formed on an underlying seed layer **141**, comprised of a metal, such as titanium, copper, gold, or the like, an alloy of two or more such materials, or any suitable combination of metals. The electrical bump **126** is disposed in contact with the UBM layer **124** or in contact with the second portion **140** of the RDL trace **120** if the UBM layer **124** is not formed. In one or more embodiments, the electrical bump **126** fills a cavity defined by the inner sidewalls and the inner bottom surface of the UBM layer **124**. The electrical bump **126** further contacts portion(s) of the UBM layer **124** extending above/over and contacting the top surface **237** (FIG. **5**) of the second re-passivation layer **122**. The electrical bump **126** comprises an electrically conductive material, such as copper, solder, aluminum, gold, nickel, silver, palladium, tin, or the like, an alloy of two or more such materials, or any suitable combination of electrically conductive materials.

(20) In the configuration shown in FIG. **1**, the electrical bump **126** is electrically coupled to the electrically conductive pad **114** by the RDL trace **120**. Because the non-planar configuration **108** of the RDL trace **120** is disposed between the signal input (i.e., the electrically conductive pad **114** and via **138**) and the signal output (e.g., electrical bump **126**) of the semiconductor device package **104**, the non-planar configuration **108** provides increased stress buffering and relaxation for the RDL electrical interconnections compared to a conventional planar (straight) RDL trace configuration regardless of whether a single or multiple RDL traces are implemented. Moreover, the non-planar configuration of the RDL trace **120** increases the contact area between the RDL trace and underlying/overlying passivation layers compared to the conventional planar RDL trace configuration, which provides for improved interlayer adhesion between the RDL trace **120** and underlying/overlying re-passivation layers **118** and **122**. The increased contact area further increases the electroplating charging path when forming the RDL trace **120** compared to the conventional planar configuration, which improves plating thickness uniformity.

(21) FIG. **2** to FIG. **6** together illustrate a method **200** for fabricating a semiconductor device (e.g., an integrated circuit) package implementing a non-planar configuration for an RDL trace(s) in accordance with at least one embodiment of the present disclosure. For ease of illustration, the method **200** is described in the example context of fabrication of the semiconductor device package **104** of FIG. **1** but is not limited to this example description.

(22) The method **200** begins at block **202** with the fabrication of an initial workpiece **201** by forming the passivation (or adhesion) layer(s) **116** on and in contact with a top surface **128** of the semiconductor device **110**. The passivation layer **116** is also formed over and in contact with at least a portion of the electrically conductive pad(s) **114**. In one or more embodiments, the passivation layer **116** is formed by depositing a passivation material using chemical vapor deposition (CVD), plasma enhanced CVD (PECVD), atmospheric pressure chemical vapor deposition (APCVD), low pressure chemical vapor deposition (LPCVD), laser assisted chemical vapor deposition (LACVD), photo assisted chemical vapor deposition (PACVD), or the like. The passivation material includes, for example, nitride, oxide, PBO, BCB, PI, or any other suitable passivation material. Suitable patterning, developing, and etching processes can then be performed to pattern the passivation material to define the passivation layer **116**. These processes (or additional processes) also form an opening in the passivation layer **116** to expose at least a portion of a top surface **205** of the electrically conductive pad **114**.

(23) At block **204**, the first re-passivation layer **118** is formed over the electrically conductive pad **114** and the passivation layer **116**, thereby forming modified workpiece **207**. The first re-

passivation layer **118** contacts the top surface **205** of the passivation layer **116** and at least the top surface **203** of the electrically conductive pad **114**. In one or more embodiments, the first re-passivation layer **118** is formed by depositing a passivation material using, for example, CVD, PECVD, APCVD, LPCVD, LACVD, PACVD, spin coating, lamination, or the like over the initial workpiece **201**. The passivation material includes, for example, passivation layer **116** comprises a material such as nitride, oxide, polymer-based materials (PI, PBO, BCB, or the like), epoxy-based materials, or any other suitable insulating passivation material.

(24) Turning to FIG. 3, the method **200** continues at block **206** with one or more lithography processes to form a hilling topography **209** in the first re-passivation layer **118**, resulting in modified workpiece **211**. In one example, the hilling topography **209** is formed using a gradient/grayscale lithography process. In this example, the first re-passivation layer **118** comprises an ultraviolet (UV) light curable material that is exposed using a photomask **213** having a gradient/grayscale design, which is represented by the varying white, black, and gradient patterns in the photomask **213**. The grayscale design of the photomask **213** allows different intensities/levels of ultraviolet (UV) light **215** to pass through the photomask **213**, thereby generating areas of different crosslink densities in the first re-passivation layer **118**. For example, a black area of the photomask **213** indicates this portion of the photomask **213** is completely shielding, i.e., 0% of UV light **215** passes through the photomask **213**. A white area of the photomask **213** indicates this portion of the photomask **213** is completely transparent, i.e., 100% of UV light **215** passes through the photomask **213**. A lighter patterned area of the photomask **213** allows more UV light **215** to pass through the photomask **213** than a darker patterned area. After exposure, a developing process is performed, and the exposed areas of the first re-passivation layer **118** with a higher cross-link density have no or less material removed compared to areas with lower cross-link density, or vice versa, depending on if the photomask **213** comprises a negative tone material or a positive tone material. The result of this process is the hilling topography **209** comprising the peaks **132** and valleys **134** described above with respect to FIG. 1. Also, the lithography process also forms a trench/opening **217** in the first re-passivation layer **118** that exposes at least a portion of the electrically conductive pad's top surface **203**. The developed portions of the photomask **213** are then cured using a baking process.

(25) In another example, a UV curable photoresist is formed over the first re-passivation layer **118** and the hilling topography **209** is first formed in the photoresist using the photomask **213**. The hilling topography **209** is then formed in the first re-passivation layer **118** using, for example, reactive ion etching (RIE). In yet another example, a multiple exposure patterning process **219**, which uses two or more photolithographic sub-processes, is performed to form the hilling topography **209** in the first re-passivation layer **118**.

(26) Turning to FIG. 4, the method **200** continues at block **208** with one or more deposition processes to form an RDL seed layer **121**, resulting in modified workpiece **221**. For example, the RDL seed layer **121** is formed on the first re-passivation layer **118** and in the trench **217**. The RDL seed layer **121**, in one or more embodiments, conforms to and contacts a top surface **225** of the first re-passivation layer **118**, sidewalls of the trench **217**, and the exposed top surface **203** of the electrically conductive pad **114**. As such, the RDL seed layer **121** has the non-planar configuration **108** described above with respect to FIG. 1. In one or more embodiments, the RDL seed layer **121** is a metal layer and is comprised of a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. PVD or another process can be used to deposit an electrically conductive material to form the seed layer **121**, such as titanium, copper, gold, or the like, an alloy of two or more such materials, or any suitable combination of electrically conductive materials.

(27) At block **210**, a mask **227**, such as a photoresist layer, is formed and patterned on the RDL seed layer **121** in accordance with a desired redistribution pattern, resulting in modified workpiece **229**. Patterning is performed to form openings through the mask **227** to expose the seed layer **121**.

At block **212**, the RDL trace **120** is formed, resulting in modified workpiece **231**. For example, an electrically conductive material is formed in the openings of the mask **227** and on the exposed portions of the RDL seed layer **121**. A process such as plating, e.g., electroplating or electroless plating, or another process(es) is performed to form the electrically conductive material. In one or more embodiments, the electrically conductive material comprises a metal, such as copper, titanium, tungsten, aluminum, or the like, an alloy of two or more such materials, or any suitable combination of electrically conductive materials. The RDL trace **120**, in one or more embodiments, comprises both the seed layer **121** and the electrically conductive material deposited thereon. The first portion **136** of the RDL trace **120** fills the trench **217**, thereby forming the via **138** in contact with the electrically conductive pad **114**. The second portion **140** of the RDL trace **120**, which is disposed opposite the first portion **136**, is configured to receive a subsequently formed UBM layer **124** and electrical bump **126**. The third portion **142** of the RDL trace **120** is disposed between the first portion **136** and the second portion **140** of the RDL trace **120** and comprises the non-planar configuration **108**. As described above with respect to FIG. **1**, the non-planar configuration **108** comprises peaks **144** and valleys **146** forming a “zig-zag” or “folding” pattern.

(28) Turning to FIG. **5**, the method **200** continues at block **214** with the removal of the mask **227** and portions of the RDL seed layer **121** on which the electrically conductive material is not formed, resulting in modified workpiece **233**. The mask **227** is removed using an acceptable process, such as an ashing process or stripping process. Once the mask **227** is removed, the exposed portions of the RDL seed layer **121** are removed using an acceptable process, such as an etching process include wet etching, dry etching, or the like.

(29) At block **216**, the second re-passivation layer **122** is formed over and in contact with the RDL trace **120** and any exposed surface of the first re-passivation layer **118**, resulting in modified workpiece **235**. In one or more embodiments, the second re-passivation layer **122** is formed by depositing a passivation material using, for example, CVD, PECVD, APCVD, LPCVD, LACVD, PACVD, spin coating, lamination, or the like over the initial workpiece **201**. The passivation material includes, for example, a material such as such as nitride, oxide, polymer-based materials (PI, PBO, BCB, or the like), epoxy-based materials, or any other suitable insulating passivation material. The second re-passivation layer **122** comprises the same or different material than the first re-passivation layer **118**. A planarization process, such as chemical mechanical polish (CMP), can be performed to planarize the top surface **237** of the second re-passivation layer **122**.

(30) At block **218**, a trench/opening **239** is formed the second re-passivation layer **122** and a UBM seed layer **241** is formed in the trench **239** and on the top surface **243** of the second re-passivation layer **122**, resulting in modified workpiece **245**. In one or more embodiments, the trench **239** in the second re-passivation layer **122** is formed by forming a photoresist (not shown) over the top surface **243** of the second re-passivation layer **122**, patterning the photoresist, and etching the second re-passivation layer **122** through the patterned photoresist using a suitable etching process (e.g., a wet etching process, a dry etching process, or the like. The trench exposes a top surface **247** of the second portion **140** of the RDL trace **120**. In one or more embodiments, the UBM seed layer **241** is a metal layer and is comprised of a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. PVD or another process can be used to deposit an electrically conductive material to form the UBM seed layer **241**, such as titanium, copper, gold, a combination thereof, or the like. The UBM seed layer **241**, in one or more embodiments, conforms to and contacts the top surface **243** of the second re-passivation layer **122**, sidewalls of the trench **239**, and the exposed top surface **247** of the second portion **140** of the RDL trace **120**.

(31) Turning to FIG. **6**, the method **200** continues at block **220** with the forming of the final UBM layer **124**, resulting in modified workpiece **249**. For example, a photoresist layer/mask **251**, is formed and patterned on the UBM seed layer **241** in accordance with a desired redistribution pattern. Patterning is performed to form openings through the mask **253** to expose the UBM seed layer **241**. An electrically conductive material is then formed in the openings of the mask **251** and

on the exposed portions of the seed layer **241** to form the final UBM layer **124** in contact with a portion of the second re-passivation layer's layer top surface **143**, sidewalls of the trench **239**, and the exposed top surface **247** of the second portion **140** of the RDL trace **120**. The UBM seed layer **241**, in one or more embodiments, comprises both the seed layer **141** and the electrically conductive material deposited thereon.

(32) At block **222**, the mask **251** and portions of the UBM seed layer **241** on which the UBM material is not formed are removed, resulting in modified workpiece **255**. The mask **251** is removed using an acceptable process, such as an ashing or stripping process. Once the mask **251** is removed, the exposed portions of the UBM seed layer **241** are removed using an acceptable process, such as an etching process include wet etching, dry etching, or the like. At block **224**, a bumping process is performed to form the electrical bump(s) **126**, resulting in the semiconductor device package **104** described above with respect to FIG. **1**. For example, an electrical bump material is deposited over the second re-passivation layer **122** and the UBM layer **124**. The electrical bump material comprises, for example, solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. The electrical bump material is reflowed into a spherical shape using a high-temperature reflow process to form the electrical bump. The electrical bump **126** is formed in electrical contact with the second portion **140** of the RDL trace **120** through the UBM layer **124** or, if the UBM layer **124** is not formed, the electrical bump **126** is formed in physical contact with the second portion **140** of the RDL trace **120**. As such, the non-planar configuration **108** of the RDL trace **120** is disposed between the signal input (i.e., the electrically conductive pad **114** and via **138**) and the signal output (e.g., electrical bump **126**) of the semiconductor device package **104**, which provides increased stress buffering and relaxation for the RDL electrical interconnections compared to a conventional planar (straight) RDL configuration.

(33) Note that not all of the activities or elements described above in the general description are required, that a portion of a specific activity or device may not be required, and that one or more further activities may be performed, or elements included, in addition to those described. Still further, the order in which activities are listed are not necessarily the order in which they are performed. Also, the concepts have been described with reference to specific embodiments. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present disclosure.

(34) Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims. Moreover, the particular embodiments disclosed above are illustrative only, as the disclosed subject matter may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. No limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope of the disclosed subject matter. Accordingly, the protection sought herein is as set forth in the claims below.

Claims

1. A semiconductor device package comprising: a semiconductor chip; a first re-passivation layer disposed over a surface of the semiconductor device chip, wherein the first re-passivation layer

includes an opening that exposes a top surface of an electrically conductive pad, and the first re-passivation layer further includes a non-planar portion relative to the surface of the semiconductor chip adjacent to the opening; and a redistribution layer (RDL) trace disposed in conformal contact with the first re-passivation layer, wherein the RDL trace comprises a non-planar configuration relative to the surface of the semiconductor chip adjacent to the opening, wherein the non-planar configuration comprises a first area of the RDL trace having a first height and a second area of the RDL trace having a second height different than the first height.

2. The semiconductor device package of claim 1, wherein the non-planar configuration comprises at least one peak and at least one valley in the RDL trace.

3. The semiconductor device package of claim 1, wherein the semiconductor device package further comprises: the electrically conductive pad disposed below the first passivation layer and in contact with the semiconductor chip and the RDL trace; a second re-passivation layer disposed above and in contact with the RDL trace; and an electrical connector disposed above and electrically coupled to the RDL trace.

4. The semiconductor device package of claim 3, further comprising: a metallization layer disposed between the RDL trace and the electrical connector, wherein a first surface of the metallization layer contacts the RDL trace, and a second surface of the metallization layer contacts the electrical connector.

5. The semiconductor device package of claim 3, further comprising: a passivation layer in contact with the surface of the semiconductor device and a bottom surface of the first re-passivation layer, and further in contact with the electrically conductive pad.

6. The semiconductor device package of claim 3, wherein the RDL trace comprises a first portion in contact with the electrically conductive pad, a second portion in contact with the electrical connector, and a third portion comprising the non-planar configuration, wherein the third portion is disposed between the first portion and the second portion of the RDL trace.

7. The semiconductor device package of claim 6, wherein a first surface of the third portion of the RDL trace transitioning from the first area of the RDL trace having the first height to the second area of the RDL trace having the second height is sloped such that a first obtuse angle is formed between the first area and the second area, and wherein a second surface of the third portion of the RDL trace transitioning from the second area to a third area of the RDL trace having a third height is sloped such that a second obtuse angle is formed between the second area and third area.

8. The semiconductor device package of claim 1, wherein the non-planar portion of the first re-passivation layer includes at least one peak having a first height above a bottom surface of the first re-passivation layer and at least one valley having a second height above the bottom surface of the first re-passivation layer, wherein the first height is greater than the second height.

9. The semiconductor device package of claim 8, wherein a range of a height difference between the first height of the at least one peak and the second height of the at least one valley is at least 2 micrometers.

10. The semiconductor device package of claim 8, wherein a range of a height difference between the first height of the at least one peak and the second height of the at least one valley is at least 5 micrometers.

11. The semiconductor device package of claim 8, wherein a thickness of the first re-passivation layer at the at least one valley is at least 5 micrometers.

12. The semiconductor device package of claim 8, wherein the non-planar portion of the first re-passivation layer includes multiple peaks having the first height above the bottom surface of the first re-passivation layer and multiple valleys having the second height above the bottom surface of the first re-passivation layer.

13. The semiconductor device package of claim 8, wherein the non-planar configuration of the RDL trace has a pattern along a plane orthogonal to the surface of the semiconductor chip that is selected from a group consisting of a hilling topography, a zig-zag pattern, a folding pattern, an

undulating pattern, and a quasi-sinusoidal pattern.

14. A semiconductor device package comprising: a semiconductor chip; an electrically conductive pad disposed in contact with the semiconductor chip; a redistribution layer (RDL) formed over the electrically conductive pad; and an electrical connector disposed over and electrically coupled to the RDL, wherein the RDL comprises: a first re-passivation layer disposed over a surface of the semiconductor chip, wherein the first re-passivation layer includes an opening that exposes a top surface of the electrically conductive pad, and the first re-passivation layer further includes a non-planar portion relative to the surface of the semiconductor chip between the opening and the electrical connector; and an RDL trace comprising a first portion in contact with the electrically conductive pad, a second portion in contact with one of the electrical connector or an underlying metallization layer in contact with the electrical connector, and a third portion in conformal contact with the first re-passivation layer so that the RDL trace has a non-planar configuration relative to the surface of the semiconductor chip between the opening and the electrical connector.

15. The semiconductor device package of claim 14, wherein the non-planar configuration comprises at least one peak and at least one valley in the RDL trace.

16. The semiconductor device package of claim 15, wherein the at least one peak is disposed in at least a first area of the third portion of the RDL trace having a first height and the at least one valley is disposed in at least a second area of the third portion of the RDL trace having a second height different than the first height, and wherein a surface of the RDL trace transitioning from the first area to the second area is sloped such that an obtuse angle is formed between the first area and the second area.

17. A method of fabricating a semiconductor device package, comprising: forming a first re-passivation layer over a surface of an underlying semiconductor chip device; patterning the first re-passivation layer, wherein the patterning forms an opening that exposes a top surface of an electrically conductive pad and a first non-planar and undulating configuration in a portion of the first re-passivation layer relative to the surface of the semiconductor chip adjacent to the opening; and forming a redistribution layer (RDL) trace in contact with the patterned first re-passivation layer, wherein the RDL trace comprises a second non-planar and undulating configuration relative to the surface of the semiconductor chip adjacent to the opening and conforming to the first non-planar and undulating configuration.

18. The method of claim 17, wherein the first non-planar and undulating configuration and the second non-planar and undulating configuration each comprise at least one peak and at least one valley.

19. The method of claim 17, wherein the first re-passivation layer is patterned using at least one of a grayscale mask or a multi-exposure process.

20. The method of claim 17, wherein patterning the first re-passivation layer forms the opening in the first re-passivation layer that exposes the electrically conductive pad, and wherein the electrically conductive pad is in contact with the semiconductor chip.
